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*ms*CNN-PLM-FAD: Enhanced predicting FAD binding sites by using protein language representation combined with multiple separable windows convolutional neural networks

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ABSTRACT

The FAD binding sites classification problem is crucial because it is directly related to many diseases, such as flavoprotein-associated diseases, developmental disorders, digestive and lipid metabolism abnormalities, anemia, cancer, cardiovascular diseases, and neurological diseases. Researchers have conducted numerous studies to classify FAD binding sites, achieving varying levels of effectiveness. Nevertheless, significant potential for enhancement persists. Our study proposes msCNN-PLM-FAD, an innovative computational model predicting FAD binding sites in electron transport proteins using a sliding window feature extraction technique and pretrained protein language models (PLMs). Our approach combines a convolutional neural network-based window scanning technique able to capture various sequence features relevant to FAD binding sites with embeddings from the ESM PLM. Built on a dataset of 12,850 samples, the model outperformed previous studies, particularly in class-balance metrics, with an area under the curve (AUC) of 0.9614 and a Matthews correlation coefficient (MCC) of 0.7491, specificity of 0.9836, and accuracy of 0.9795, while maintaining a sensitivity of 0.8704. The results of this study demonstrate the potential to improve the accuracy of FAD binding site prediction, thereby promoting the development of drugs for diseases associated with FAD deficiency as well as therapeutic approaches that target bacterial FAD synthesis without affecting human health.

1. Introduction

Energy is an essential resource for all forms of life on this planet. Flavin Adenine Dinucleotide (FAD), along with NAD, plays an important role in cellular respiration by transporting electrons from glucose to produce ATP. This is the cell's main source of energy, produced through glycolysis, the Krebs cycle, and the electron transport chain. The important role of FAD in the electron transport chain is demonstrated through Fig. 1, further clarifying the role of FAD in overall human health. Numerous studies concerning human health issues confirmed the importance of FAD. Flavoprotein, a protein containing the flavin group, is connected with numerous disorders in humans, with two-thirds of human flavoprotein linked to these conditions [1]. Furthermore, riboflavin (vitamin B2) insufficient can result in insufficient flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN), two essential coenzymes in many metabolic activities in the body. When these substances are deficient, the body can experience a variety of health issues, including developmental disorders, abnormalities in the digestive system and fat breakdown, anaemia, neurological diseases caused by enzyme dysfunction, an increased risk of cancer and cardiovascular disease, migraines, vision loss, and skin lesions [2]. Additionally, FAD has demonstrated its value in drug design. A particular metabolic protein that employs FAD is crucial for bacterial pathogenicity. The synthesis of FAD or the production of FAD analogues represents a potential target for the inhibition of bacterial infections [3]. Another study indicates that the FAD synthesis pathways in humans and bacteria involve different enzymes. This property enables the design of drugs that specifically target the bacterial FAD synthesis pathway, leaving human FAD synthesis enzymes unaffected [4].

Researchers including those in the field of computer science have recognized the importance of using the power of computer algorithms to assist in predicting FAD binding sites to help improve the understanding

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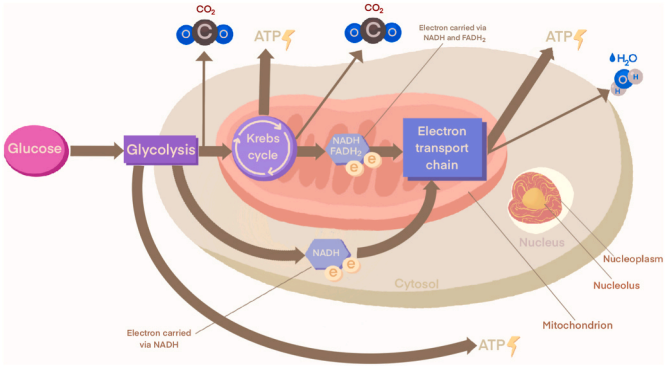


Fig. 1. The cellular respiration mechanism utilizes the electron transport chain to keep and transmit electrons.

of electron transfer in the electron transport chain. However, the complexity of protein sequences has always been a limitation that has prevented the implementation of the above problem from achieving very high results. This is largely due to the fact that current protein feature extraction methods such as the position-specific scoring matrix (PSSM) were not really effective and past machine learning algorithms still had many limitations when trying to learn patterns from this type of data. Mishra and Raghava [5] utilized support vector machine (SVM) learning to predict FAD binding sites and established a complimentary web server to support their research. However, the results are still modest with an accuracy of 70.31% and an MCC of 0.41. The Fang, Noguchi, Yamana group [6] enhanced prediction performance by integrating evolutionary information. Afterwards, Quang-Thai Ho et al. [7] enhanced the identification of FAD binding sites in electron transport proteins by integrating data from the extraction and analysis of word representations derived from pre-trained BERT large language models with biochemical properties informed by the Amino Acid Index and PSSM. The data was subsequently input into three classifiers (k nearest neighbours, random forest, and 2D CNN) for comparative analysis, yielding an 11% enhancement in accuracy relative to the previous approach. The study showed positive results, demonstrating a sensitivity of 0.8519, a specificity of 0.8562, and an overall accuracy of 0.8560, indicating a good classification capability. Nonetheless, an important concern in biological analyses is that the Matthews correlation coefficient (MCC) is at 0.35, signifying an inadequate balance across classes. Despite the high sensitivity value, the low MCC suggests that the model may insufficiently forecast false negative situations, hence significantly weakening its reliability in real applications. In biology, researchers often try to minimize both false positives and false negatives. Jian Wu et al. [8] mentioned that when biological studies attempt to gain a comprehensive understanding of a particular problem, false positives and false negatives are a significant problem. Therefore, enhancing accuracy to reduce both false positives and false negatives remains a critical concern. Although some progress has been made in previous studies on FAD location classification, the modest recognition rates make them unreliable enough to be applied to practical applications related to human health. These limitations hinder their potential as a robust foundation for advancing research aimed at improving human health outcomes related to FAD.

In general, the methods used to solve the FAD site classification problem are divided into two stages, feature extraction and algorithm classification. For the first stage, past methods for extracting features from protein sequences in general and positions in protein sequences in particular, such as PSSM, often operate on the assumption that positions in the active sequence are independent, based only on the frequency of amino acids without considering the protein structure. Thus, many important features of the sequence are lost, time-consuming and unable to sufficiently represent context-specific and sequential dependencies. Protein feature capture has taken a new step forward with the advent of protein language models (PLMs) thanks to the significant advances made by natural language models (NLPs), especially after the introduction of the Transformer architecture proposed by Vaswani et al. [9]. As demonstrated by B. Victorri [10], both natural languages and protein sequences share many characteristics in their presentation and representation, so it is possible to view protein sequences as a special type of biological language. Deep learning models have exhibited significant potential in the analysis and prediction of proteins' biological properties compared to traditional methods. Rives et al. [11] demonstrated that a language model can be trained on 250 million protein sequences, enabling it to autonomously learn important biological features, including secondary structures and evolutionary relationships among protein families. Embeddings generated by PLMs maintain the sequential structure of the input while encoding high-level contextual features. Moreover, the application of language models to biological data has been extensively investigated in recent studies, leading to notable advancements across various tasks [12–15].

In the stage of applying classification algorithms to processed data, previous studies primarily relied on classical methods such as SVM, kNN, and 2D CNN. Although these methods achieved moderate performance, they are not suitable for handling long, sequential protein data, particularly in capturing relationships between adjacent positions in the protein chain. Also, the relationships between adjacent amino acid ranges vary in each case, and the aforementioned methods do not fully capture these connections. To address the limitation in capturing the diverse relationships between adjacent amino acid ranges and to better suit sequential data, Quang Thai Ho et al. [16] applied multiple windows scanning techniques in convolutional neural networks (mCNN) on the binary classification problem of whole protein sequences and achieved impressive improvements over the previous method on all metrics, typically with MCC improved from 0.6383 to 0.8942. Furthermore, several studies have successfully extended this methodology to a wide range of biological data analysis tasks, consistently reporting significant performance improvements across different domains [17–19]. However, this method has the limitation that the number of model parameters increases significantly due to the use of multiple convolutional windows, making the model suitable for low-dimensional data such as PSSM, but with high-dimensional embedding data such as Lin's 15B largest model with 5120 dimensions [20], this method can face many challenges in terms of performance and computational cost.

To address the limitations of previous studies on FAD binding site classification that still have many limitations in the ability to accurately classify FAD locations due to limitations in the implementation method, our study proposes the msCNN-PLM-FAD model to create a method that achieves higher accuracy than previous FAD location recognition methods by combining PLMs with multiple windows scanning techniques in convolutional neural networks, while optimizing the computational cost commonly encountered when combining high-dimensional embedding data with multi-window models by incorporating separable convolutional architectures to reduce parameter count and computational complexity without compromising representational capacity.

2. Materials and methods

This research aims to improve the performance of predicting FAD binding sites in electron transport proteins. *ms*CNN-PLM-FAD consists of two primary components: a PLM-based feature extractor with sliding windows and a CNN-based classifier. The pre-trained language models can extract vast amounts of information on biological language by interpreting an entire protein sequence as a sentence and its components, amino acids, as individual words. Following this procedure, using a sliding window to evaluate the impact of neighboring amino acids on the discrimination between 2 sites, combining with a CNN-based deep neural network model extracted the pattern of these biological functions and executed binary classifications for the FAD binding sites and non-FAD sites task. Fig. 2 illustrates the workflow for predicting FAD

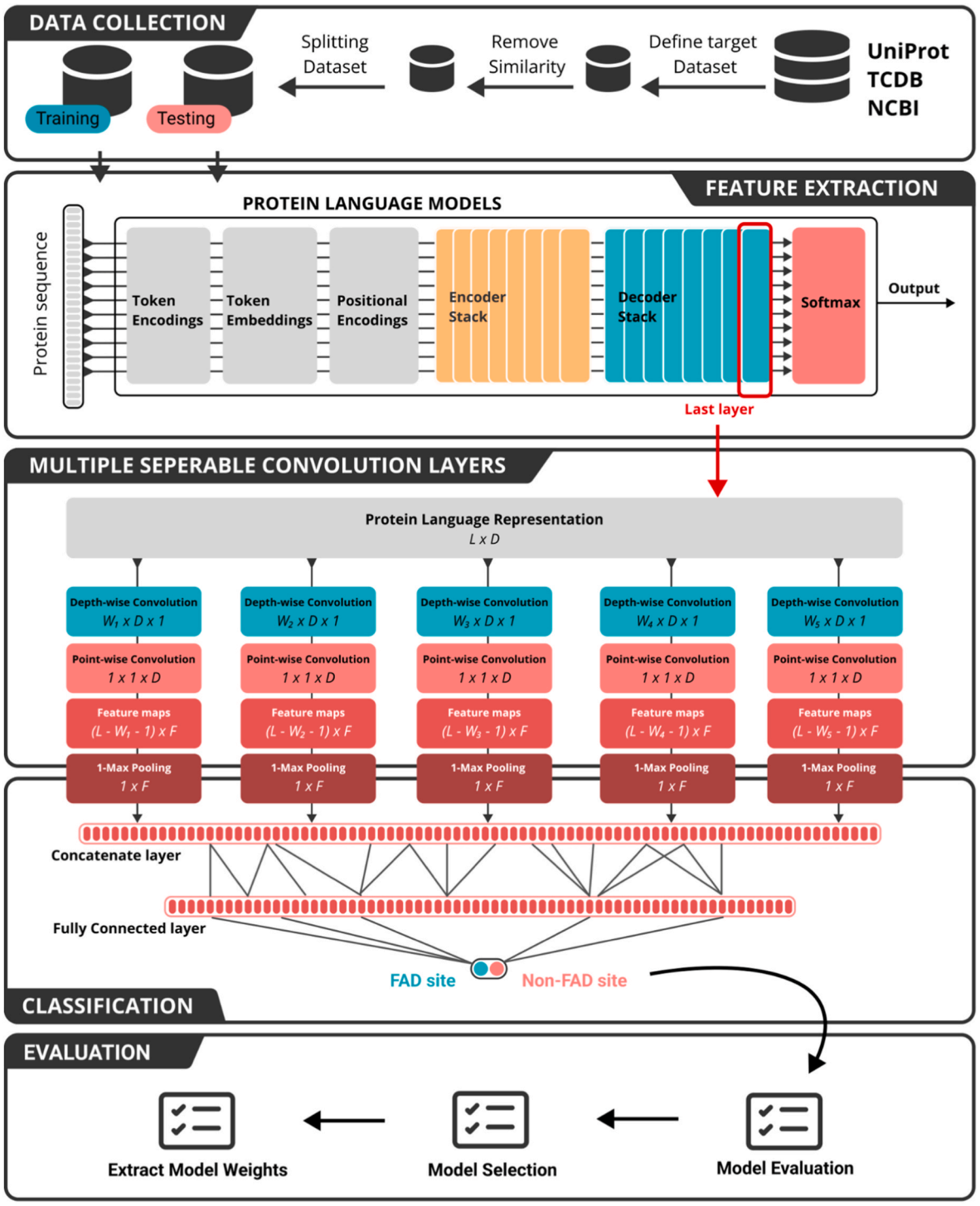


Fig. 2. The workflow for building FAD binding site prediction models.

binding sites in this study, involving four stages: data collection, feature extraction, classification, and evaluation. The details for each component are outlined in the following sections.

2.1. Data collection

Based on the dataset collected by Quang-Thai Ho et al. [7], to correspond with standard procedures, we re-examined our issue and leveraged the advancements in sequence-based protein recognition facilitated by PLMs. The dataset used in this study was collected from the Universal Protein (UniProt) database [21]. A total of 22,506 data samples were initially collected, then reduced to 12,850 after applying BLAST [22,23] to eliminate those with a sequence identity exceeding 30% from the dataset. The samples were then divided into a training set

Table 1
Statistics of the survey dataset.

	Original	Similarity <30%	Training	Independent
FAD binding sites	523	320	266	54
Non-FAD binding sites	21,983	12,530	11,070	1460
Total	**22,506**	**12,850**	**11,336**	**1514**

and a testing set with the numbers of 11,336 and 1,514, respectively. Table 1 presents the dataset statistics, whereas Table 2 offers the accession numbers of all surveyed samples.

2.2. Protein language representation

In this work, we utilized modern methods of PLM models on our dataset. In order to effectively capture those complex sequence patterns and structural information essential for predicting FAD binding sites, we make use of PLMs from the ProtTrans series and ESM variants. Fig. 3 illustrates the process of extracting embeddings from a protein sequence using PLM. PLMs serve as a robust tool based on the pretraining concept from natural language processing, utilized for the analysis of protein sequences. The models are subjected to unsupervised pre-training using extensive protein sequence data sets, allowing them to gather structural and functional insights from the protein sequences. This enables the generation of high-dimensional embeddings that reflect the complex variances and contextual relationships of the sequences. The next sections will provide insight into these technical mechanisms.

Similar to language models like BERT [24] for natural language processing, the term "ProtTrans" refers to a collection of pre-trained language models created especially for protein sequences. The models were built on data from UniRef and BFD containing up to 393 billion amino acids and powered by the Summit supercomputer using 5616 GPUs and TPU Pod up-to 1024 cores. The trained models showed that protein embeddings from unlabeled data capture some important biophysical features of protein sequences. Elnaggar et al. [25] published their models as open source, accessible at https://github.com/agemagician/ProtTrans. In this study, we use embeddings from two representative language models, ProtT5-XL and ProtBERT. These two models were chosen based on their outstanding performance in previous molecular biology tasks, as well as differences in architecture (auto-regressive and auto-encoder), which provide a multi-dimensional view of the representational capabilities of protein embeddings.

ESMs are a family of pretrained models similar to NLPs but tuned to process protein sequences instead of natural language, born with the goal of modeling protein language at the evolutionary scale. Rives et al. [11] used unsupervised learning to train the first version, ESM-1, a deep contextual language model on 250 million protein sequences. Lin et al. [20] then further improved the unsupervised trained ESM-2 models to 15 billion parameters, demonstrating the ability to learn information that allows prediction of the protein structure at the atomic level, revealing new insights into the range and diversity of natural proteins and aiding the discovery of novel protein structures and functions. The performance of ESM-2 is significantly improved over its predecessor, with low parameter versions such as ESM-2 150 M still tending to capture the structure more accurately than the higher parameter count ESM-1 650 M. All models are publicly available at Github page https://github.com/facebookresearch/esm. In this study, we use the ESM-2 650 M model with 33 transformer layers and 650 million parameters, which is the average size in the ESM-2 family based on the balance between prediction performance and computational cost.

2.3. msCNN - multiple separable windows convolutional neural networks

To address the computational cost, particularly in settings involving high-dimensional embeddings such as those from pretrained protein language models, we incorporate separable convolutional filters—specifically using a combination of depthwise and pointwise convolutions. As shown in Fig. 4, depthwise separable convolutions and standard convolutional layers differ fundamentally in how they function. Conventional convolutions combine spatial and channel-wise information in a computationally demanding procedure by applying filters simultaneously across all spatial dimensions and input channels. Depthwise separable convolutions, on the other hand, do this in two steps: a depthwise convolution applies a single filter to each input channel separately, and a pointwise convolution uses 1×1 filters to combine channels. With this approach, both the computational cost and the amount of memory that is used are significantly reduced, which enables faster training and inference. The concept was elucidated by Chollet [26], which interprets depthwise separable convolutions as a specialized form of Inception modules with a maximally large number of towers. The proposed Xception architecture, replacing Inception modules with depthwise separable convolutions, achieves slightly better performance than Inception V3 on ImageNet and significantly outperforms it on a larger dataset of 350 million images, with the same number of parameters, indicating more efficient parameter usage. Similarly, Howard et al. [27] use depthwise separable convolutions to produce models that are small and fast for mobile and embedded devices. MobileNet's streamlined architecture, which can be adjusted using hyper-parameters to balance latency and accuracy, works well for tasks like object detection and fine-grained classification. Both works underscore the efficiency of depthwise separable convolutions, though their reduced channel interactions require careful architectural design to preserve model performance. This architectural selection, executed within the *m*CNN framework, significantly reduces the number of trainable parameters while maintaining the model's capacity to identify prominent features across various receptive fields. This allows us to integrate rich protein embeddings effectively without the heavy computational cost that often limits deep CNN-based methods. We propose the *ms*CNN-PLM-FAD model to solve the binary classification problem of FAD binding sites on protein sequences, promising to improve the overall performance.

Table 2
Accession numbers of all proteins used in this study.

Training dataset					Independent dataset
P00455	P13804	P38038	Q03304	Q96HE7	P0DPQ8
P06149	P16083	P55931	Q06278	Q9UI17	C6KT68
P08165	P16603	P83223	Q39243	Q9YHT1	O67931
P09622	P17412	Q03103	Q687X5	Q9Z0X1	Q9AL95

2.4. Sliding windows with multiple windows scanning techniques in convolutional neural networks

We performed experiments with various sliding window sizes to evaluate the impact of neighboring amino acids on the discrimination between FAD binding sites and non-FAD sites, aiming to identify the optimal size for this purpose. Using a sliding window across the protein chain, together with neighboring amino acids, helps understanding of their impact on the target amino acid location.

The highlight of our approach in this paper is the application of multi-window separable convolutional filters to determine the labels of sites in the sequence using sliding windows as input. The approach employing multi-window convolutional filters has demonstrated effectiveness for the entire protein sequence topic in prior research [16]. This study utilizes the aforementioned strategy to discover FAD binding sites within the sequence, while simultaneously minimizing computational demands by decreasing the number of parameters through the utilization of Separable Convolutional Filters, without sacrificing model accuracy. Separable convolutions reduce the number of parameters by separating the convolution process into two stages: depthwise convolution and pointwise convolution. Specifically, depthwise convolution applies an individual 2D filter for each input channel, subsequently utilizing pointwise convolution to merge the information across channels. Consequently, the quantity of parameters is markedly diminished, particularly when kernel sizes and channel counts are substantial.

We construct a *ms*CNN model to analyze the input data using multiple convolutional filters of varying window sizes. We also examine several window combinations and single window sizes to determine the optimal configuration for identifying FAD binding sites. Employing

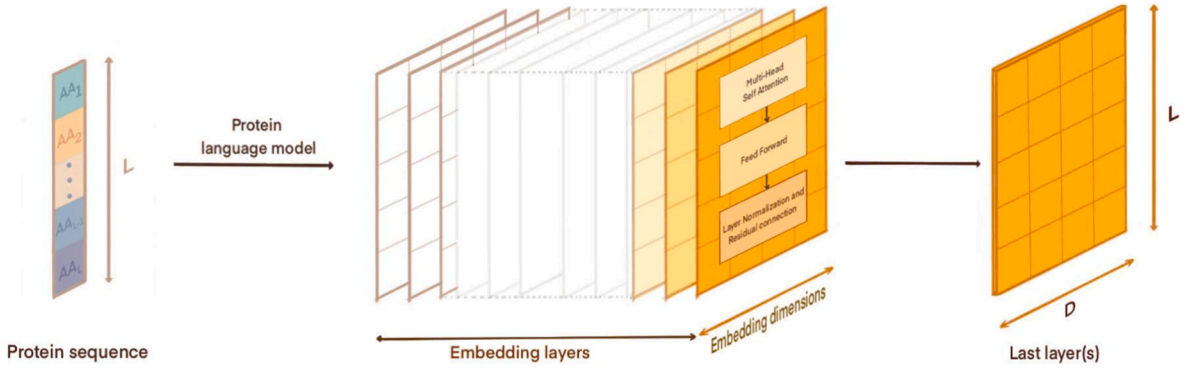


Fig. 3. The process of extracting embeddings from a protein sequence using PLM.

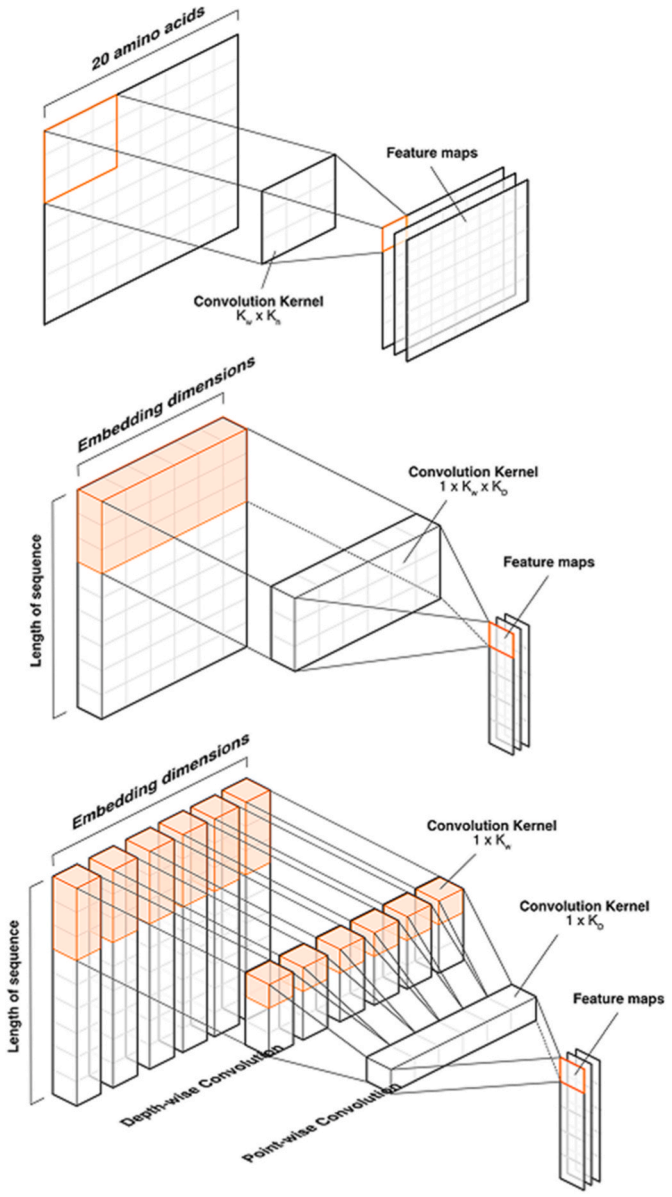


Fig. 4. Comparison of CNN, mCNN, and msCNN Architectures in protein prediction.

window combinations allows the extraction of features at various levels within the input data. To keep the most important features of every filter, a max pooling layer follows each convolutional layer. Using ReLU to increase the ability to learn non-linear characteristics, the output is then flattened and run through a fully connected layer. We also apply Dropout (0.7) before feeding the data to the output layer, where a Softmax function is used for classification. The output layer is modified by L2 regularization to enhance the model's generalization capability. We will also compare hidden nodes settings, filter sizes, sliding window lengths as well as data embeddings from different PLMs in the section below to find the optimal settings for the problem.

2.5. Performance metrics

The predictive model was trained on the training dataset, with hyperparameters optimized to enhance performance on the validation dataset. The model's performance was measured with common metrics, including sensitivity, specificity, accuracy, Matthews correlation coefficient (MCC), and area under the receiver operating characteristic curve (AUC). MCC and AUC were essential for evaluating the model's capability in accurately identifying FAD-binding and non-binding residues, respectively.

To assess the resilience of our model, we performed supplementary experiments evaluating its performance across various window combinations, hidden nodes, filter sizes, sliding windows size and feature sets, including ProtT5, ProtBert, and ESM2-650 M embeddings. The outcomes continuously indicated the superior performance of our protein PLMs-based model, where it achieved a MCC of 0.7491 and an AUC of 0.9614, strongly surpassing the previous method.

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$MCC = \frac{TP*TN - FP*FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

3. Results and discussions

3.1. Performance comparison with different single window sizes

Initially, we examined the influence of various single window sizes on model performance. Table 3 illustrates a gradual improvement in

Table 3
Performance comparison with different single window sizes.

Single window sizes	Sensitivity	Specificity	Accuracy	MCC	AUC
4	0.8519	0.9582	0.9544	0.5862	0.9520
8	0.8704	0.9664	0.9630	0.6368	0.9596
12	0.8704	**0.9719**	**0.9683**	**0.6676**	**0.9684**
16	**0.9074**	0.9548	0.9531	0.6036	0.9666

sensitivity, specificity, accuracy, AUC, and MCC values with increasing window size, peaking at window size 12 with MCC = 0.6676 and AUC = 0.9684, before declining thereafter.

3.2. Performance comparison with different window combinations

Next, we examined the impact of various window size combinations on model performance. Starting with window size 12, which was the optimal outcome in the previous experiment, we experimented with combination window sizes based on 12. Table 4 indicates that the maximum MCC (0.7321) and AUC (0.9653) values were achieved with a combination of 4, 6, 8, 10, and 12 window sizes. The results indicated that the MCC value progressively increased with the number of windows and peaked at window combinations of 4, 6, 8, 10, and 12, while the AUC values indicated little difference among the combinations. This result reflects the advantage of applying multiple windows, when each window size recognizes different and complementary biological features. This result illustrates the complexity of FAD binding sites; when a single window can only record information at a certain size, multiple windows provide diverse information at many different sizes.

3.3. Performance comparison with different hidden nodes

Next, we examine the impact of varying hidden node configurations on the model's performance. The results indicate minimal variation among the settings when evaluating the metrics. The configuration with 1024 hidden nodes shows the highest MCC value at 0.7321, indicating the most balance among the model's metrics. This is closely followed by the 512 hidden nodes setting, which yields results slightly lower than the 1024 hidden nodes configuration. However, the results in Table 5 show that the 1024 hidden nodes configuration offers better coverage overall.

3.4. Performance comparison with different filter sizes

We examined the influence of the number of convolutional filters on model performance. Table 6 illustrates our analysis of filter sizes ranging from 64 to 2048, using the optimal settings determined in the previous experiments. The filter size of 512 achieved the maximum MCC of 0.7491, accompanied by an appropriate AUC of 0.9614. Filter size 1024 also gives quite similar results to filter size 512. However, filter size 512 gives better sensitivity as well as more balanced results between metrics, achieved at a smaller filter size.

3.5. Performance comparison with different features

In this study, we represent protein sequences using popular embedding models from ProtTrans and ESM2. We use two popular ProtTrans variants, ProtT5-XL and ProtBert, with 1024 dimensions, to incorporate rich sequence features, which include basic information about amino acid sequences as well as deeper structural and functional insights. For ESM2, we choose the esm2_t33_650 M_UR50D model to compare with ProtBERT and ProtT5-XL because of the similar embedding size (1280 dimensions), which helps minimize the difference when comparing models.

Table 7 demonstrates that ESM2-650 M outperforms the other two embeddings across all metrics. Most clearly in the MCC metric, ESM2-650 M achieves MCC = 0.7491, far surpassing when compared to 0.5417 and 0.4889 achieved when using ProtT5-XL-UniRef50 and ProtBert, respectively. It can be concluded that ESM2 embeddings have superior performance when applied to the problem of FAD binding site discrimination. Fig. 5 provides a comparison of the overall performance of the three models above through the Receiver Operating Characteristic Curve.

3.6. Determining optimal sliding window length and validating model robustness

As the final step in our hyperparameter optimization process, we experimented with input datasets of varying sliding window lengths. The selected sliding window sizes for comparison are determined by the aforementioned filter window size, with the maximum size being 12. We choose window lengths of 13, 15, 17 and 19 respectively for comparative analysis.

As shown in Table 8, we performed 5-fold cross-validation for each window length to ensure statistical robustness. A distinct trend can be seen in the results, which are displayed as mean and standard deviation (SD). The mean MCC value progressively increases, peaking at a window length of 17 (MCC = 0.6559 ± 0.0754), before experiencing a minor decline at length 19. This high performance at length 17 was also consistent across other metrics, such as a mean AUC of 0.9661 ± 0.0108. Crucially, the low standard deviations observed across all metrics provide strong quantitative evidence of the model's stability and reliability, indicating that the performance is not sensitive to data partitioning. For

Table 4
Performance comparison with different window combinations.

Window sizes Combinations	Sensitivity	Specificity	Accuracy	MCC	AUC
[12]	0.8704	0.9719	0.9683	0.6676	0.9684
[12, 10]	**0.8889**	0.976	0.9729	0.7047	0.9709
[12, 10, 8]	0.8704	0.9774	0.9736	0.7028	0.9617
[12, 10, 8, 6]	**0.8889**	0.9767	0.9736	0.7093	**0.9738**
[12, 10, 8, 6, 4]	0.8333	**0.9842**	**0.9788**	**0.7321**	0.9653
[12, 10, 8, 6, 4, 2]	0.8704	0.9671	0.9636	0.6405	0.9681

Table 5
Performance comparison with different hidden nodes.

Hidden nodes	Sensitivity	Specificity	Accuracy	MCC	AUC
128	0.8704	0.9774	0.9736	0.7028	0.9606
256	0.8519	0.9808	0.9762	0.7162	0.9513
512	0.8519	0.9815	0.9769	0.7215	0.9558
1024	0.8333	**0.9842**	**0.9788**	**0.7321**	**0.9653**
2048	**0.8889**	0.9781	0.9749	0.7187	0.9589

Table 6
Performance comparison with different filter sizes.

Filter sizes	Sensitivity	Specificity	Accuracy	MCC	AUC
64	0.8704	0.9719	0.9683	0.6676	0.9544
128	0.8704	0.9767	0.9729	0.6981	**0.9694**
256	0.8704	0.9733	0.9696	0.6760	0.9660
512	0.8704	0.9836	**0.9795**	**0.7491**	0.9614
1024	0.8333	**0.9842**	0.9788	0.7321	0.9653
2048	**0.8889**	0.9753	0.9722	0.7001	0.9625

Table 7
Performance comparison with different features.

Features	Sensitivity	Specificity	Accuracy	MCC	AUC
ESM2-650M	**0.8704**	**0.9836**	**0.9795**	**0.7491**	**0.9614**
ProtT5-XL-UniRef50	0.7963	0.9554	0.9498	0.5417	0.9363
ProtBert	0.7222	0.9534	0.9451	0.4889	0.9023

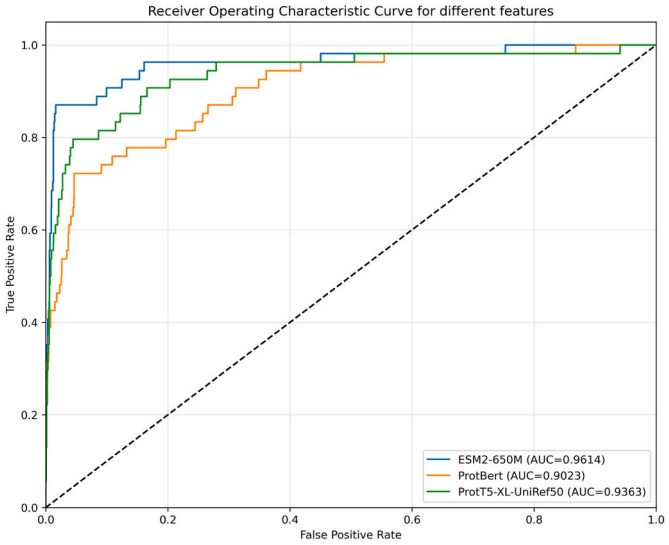


Fig. 5. ROC curve for performance with different features.

Table 8
Performance comparison with different sliding window lengths.

Window Lengths	Sensitivity	Specificity	Accuracy	MCC	AUC
Cross-validation (5-folds)					
13	0.8947 ± 0.0218	0.9780 ± 0.0063	0.9761 ± 0.0062	0.6598 ± 0.0542	0.9716 ± 0.0132
15	0.8647 ± 0.0151	0.9789 ± 0.0045	0.9763 ± 0.0047	0.6477 ± 0.0449	0.9684 ± 0.0071
17	0.8610 ± 0.0416	0.9791 ± 0.0093	0.9764 ± 0.0084	0.6559 ± 0.0754	0.9661 ± 0.0108
19	0.8724 ± 0.0373	0.9710 ± 0.0058	0.9687 ± 0.0061	0.5952 ± 0.0474	0.9678 ± 0.0120
Independent					
13	0.8333	0.9801	0.9749	0.6997	0.9638
15	**0.8704**	0.9801	0.9762	0.7224	0.9668
17	**0.8704**	**0.9836**	**0.9795**	**0.7491**	0.9614
19	**0.8704**	0.9788	0.9749	0.7124	**0.9724**

our final, optimized model, the sliding window length of 17 was chosen since it produced the most balanced results. This configuration was used for all subsequent comparisons against previous methods.

3.7. Delineating the contribution of sequence representation and model architecture

To disentangle the contribution of pretrained protein representations from that of the downstream model architecture, we conducted a controlled ablation study in which all classifiers were trained using identical residue-level embeddings derived from ESM-2. The proposed multi-scale msCNN was compared against three baseline models with progressively stronger inductive biases: Logistic Regression (linear global), a Shallow MLP (global non-linear), and a Single-window CNN (local fixed-scale).

As summarized in Table 9, architectural inductive bias plays a decisive role in determining predictive performance. Logistic Regression demonstrated a high AUC of 0.9666 and a Sensitivity of 0.9259, suggesting that ESM-2 embeddings possess considerable linear separability. Conversely, the model's precision was relatively low at 0.2732, and its MCC was 0.4749, which implies a limited ability to reduce false positives. The Shallow MLP, however, yielded an MCC of 0.6109 on the independent test set, implying that while global non-linear transformations partially contribute to the refinement of decision boundaries, they are still inadequate for the accurate localization of binding residues.

Introducing locality via a Single-window CNN further improved performance, yielding an independent test MCC of 0.6676. This gain highlights the importance of modeling local sequence context for FAD-binding site prediction. Notably, the proposed msCNN achieved the highest mean MCC across five-fold cross-validation (0.6559 ± 0.0754), outperforming the Single-window CNN with statistical significance under Welch's *t*-test ($p = 0.0466$), thereby demonstrating the advantage of multi-scale contextual modeling over a fixed receptive field.

On the independent test set, msCNN exhibited strong generalization, achieving an MCC of 0.7491 and a high specificity (0.9836). Although its sensitivity (0.8704) was marginally lower than that of the linear baseline, this reduction was accompanied by a substantial increase in precision (0.6620 versus 0.2732). This trade-off indicates that the multi-scale inductive bias effectively filters background noise, converting the representational capacity of pretrained embeddings into a more reliable and experimentally actionable predictor.

3.8. Comparative performance and statistical validation compared to prior approach

Table 10 shows the substantial enhancement in classification reliability in the binary classification problem of FAD binding sites between the proposed msCNN-PLM-FAD model and the previous work, FAD-BERT. The most critical gains are observed in the balanced metrics, which are essential for this highly imbalanced dataset. The Matthews Correlation Coefficient (MCC) more than doubled from 0.3503 (Baseline) to 0.7491 (Proposed). Our new model shows a slight increase in sensitivity (0.8704) over the previous model (0.8519). It also has a significant enhancement in specificity (0.9836) compared to the prior result (0.8562), indicating improved resistance to false positives. At the same time, the accuracy of the *ms*CNN-PLM-FAD model (0.9795) also outperforms FAD-BERT (0.8560), demonstrating a great improvement

Table 10
Performance comparison with previous method.

Models	Sensitivity	Specificity	Accuracy	MCC
***ms*CNN-PLM-FAD**	**0.8704**	**0.9836**	**0.9795**	**0.7491**
kNN (FAD-BERT)	0.7407	0.7623	0.7616	0.21
Random Forest (FAD-BERT)	0.8148	0.7938	0.7946	0.27
SVM (FAD-BERT)	0.8519	0.8514	0.8514	0.34
2D CNN (FAD-BERT)	0.8519	0.8562	0.8560	0.35

Table 9
Statistical evaluation of architectural inductive biases using 5-fold Cross-validation and generalization performance on the Independent Test Set.

Classifier	Architectural Bias	Internal Validation (5-fold CV)		Independent Test			
		5-fold CV MCC (Mean ± SD)	P-value	Sensitivity	Precision	MCC	AUC
Logistic Regression	Linear/Global	0.4890 ± 0.0597	0.0063	0.9259	0.2732	0.4749	0.9666
Shallow MLP	Non-linear/Global	0.5235 ± 0.0679	0.0247	0.8889	0.4444	0.6109	0.9473
Single-window CNN	Non-linear/Local (Fixed)	0.5566 ± 0.0472	0.0466	0.8704	0.5341	0.6676	0.9684
msCNN (Proposed)	Non-linear/Local (Multi-scale)	0.6559 ± 0.0754	-	0.8704	0.6620	0.7491	0.9614

in the class balance and overall performance reliability.

To validate the stability and statistical significance of these improvements, a Non-parametric Bootstrap analysis (K = 5000 iterations) was employed on the proposed model's predictions to generate 95% Confidence Intervals (CI) for each metric. Due to the unavailability of per-instance predictions for the FAD-BERT baseline, a standard paired statistical test, such as the *t*-test, McNemar's Test, or Paired Bootstrap, was not feasible. Therefore, we adopted a robust one-sample comparison, testing whether the fixed-point baseline metrics fell outside our model's 95% CIs. While this approach comparison is inherently less powerful than a true paired test, the sheer magnitude of the observed differences provides compelling quantitative evidence of the proposed model's superiority.

As shown in Table 11, the baseline MCC (0.3503), Accuracy (0.8560), and Specificity (0.8562) all fall starkly outside and far below the 95% CIs of our proposed model. For instance, the baseline MCC of 0.3503 is far removed from the new model's 95% CI of [0.6587, 0.8280]. Conversely, while the sensitivity point-estimate showed a slight increase (0.8704 vs 0.8519), the baseline value (0.8519) fell within the 95% CI [0.7727, 0.9535]. This confirms the change is not statistically significant. It is crucial, however, to contextualize this metric. The test set contained a low number of positive FAD-binding sites (N = 54). Despite this sample size, the model maintained, or slightly increased, the high true positive rate while simultaneously achieving a substantial and statistically significant reduction in false positives. This constitutes a critical and robust finding, demonstrating an improved precision-recall trade-off, crucial for identifying rare functional sites.

Despite the substantial gains in MCC and specificity, we explicitly acknowledge that the improvement in sensitivity (0.8704 vs 0.8519) did not reach statistical significance, as the baseline value falls within our model's 95% confidence interval. This is primarily attributed to the intrinsic sparsity of the dataset, where the independent test set contains only 54 positive FAD-binding residues. Such a small number of positive examples limits the statistical power to detect marginal changes in true positive rates. Therefore, our claim of 'balanced performance' is not predicated on a significant increase in sensitivity, but rather on the model's superior ability to suppress false positives, which is evidenced by the significant rise in specificity and MCC while successfully maintaining a high level of sensitivity. This refined performance profile is critical for practical applications, as it ensures that the identified binding sites are highly reliable for downstream experimental validation.

3.9. Improving computational efficiency over mCNN

In this study, in addition to improving the accuracy of the model in the FAD binding sites classification problem, we strive to further improve computational efficiency compared to the mCNN model. Table 12 illustrates the performance similarities between the two msCNN-PLM-FAD models that we proposed compared to the previously successful mCNN model. The comparison metrics do not reflect too much difference between the two models, while the number of parameters is reduced by 79.35%, promising to bring more effective and easier application in numerous instances.

Table 11
Bootstrap statistical comparison (K = 5000 iterations).

Metric	Baseline Value (Fixed)	Proposed Model (Mean)	Proposed Model (95% CI)	Significance
MCC	0.3503	0.7484	[0.6587, 0.8280]	Significant
Accuracy	0.856	0.9796	[0.9723, 0.9868]	Significant
Specificity	0.8562	0.9836	[0.9767, 0.9897]	Significant
Sensitivity	0.8519	0.8696	[0.7727, 0.9535]	Non-Significant

3.10. Assessing generalization capability on a novel non-redundant dataset

To mitigate potential biases arising from intra-protein residue correlations and ensure a rigorous assessment of generalization, we complemented our evaluation with a protein-centric framework. This approach evaluates performance on individual proteins rather than on a pooled residue basis, thereby providing a more transparent and realistic view of the model's utility. For this purpose, we expanded our independent evaluation to a cumulative pool of 2746 samples across seven distinct proteins (detailed in Table 13), comprising four proteins from the Standard Independent Set and three newly curated proteins (P53571, P56968, and Q16881). These novel sequences share less than 30% sequence identity with the training data, enabling a robust assessment of the model's ability to generalize across evolutionary distances.

The results are compelling: msCNN-PLM-FAD demonstrates remarkably stable discriminative power, with per-protein AUC values ranging from 0.9457 to 0.9943 (mean AUC = 0.9661). In terms of identification capacity, the model successfully recovered 91 out of 97 total FAD-binding residues (0.9381 total sensitivity). Notably, on the Newly Curated set, the model correctly identified 42 out of 43 actual binding sites, achieving good recall on unseen sequences. This consistent performance across diverse proteins, varying significantly in sequence lengths and functional contexts, indicates that the model's predictive logic is grounded in universal biochemical motifs captured by the ESM-2 embeddings and our multi-window CNN architecture, rather than specific sequence memorization.

We observed that while the AUC remains high, the Matthews Correlation Coefficient (MCC) exhibits variation across individual proteins, ranging from 0.4582 to 0.8415. This variation is a direct reflection of the intrinsic biological sparsity and extreme class imbalance characteristic of FAD-binding sites, which represent only approximately 2.5% of the total residues (320 out of 12,850 samples). In challenging targets such as P0DPQ8, where binding sites are exceptionally scarce, the MCC is highly sensitive to even minimal false positives. Despite this, the model's performance remains significantly superior to previous baseline methods (e.g., FAD-BERT's MCC of 0.35).

These protein-level results reinforce the conclusion that msCNN-PLM-FAD is a robust and stable heuristic tool for novel sequence annotation. This sequence-based generalization serves as a critical foundation for the more advanced structural analyses presented later in this study. Specifically, in the section "Quantitative Evaluation of Structural Predictive Utility," these predictions are further validated by projecting them onto 3D atomic coordinates to assess their spatial precision and enrichment (refer to Table 14).

3.11. Visualization of Feature Representations Using t-SNE and UMAP

To gain deeper insights into the feature representations learned by our proposed model for FAD binding site classification, we employed both t-distributed Stochastic Neighbor Embedding (t-SNE) [28] and Uniform Manifold Approximation and Projection (UMAP) [29] to visualize high-dimensional data in a two-dimensional space. The resulting visualizations are presented in Figs. 6 and 7, corresponding to the t-SNE and UMAP projections, respectively. Specifically, we compared the original ESM-2 embeddings (650 M model, 1280-dimensional features per residue, averaged over a sequence length of 17 residues) with the learned features extracted from the fully connected layer of the msCNN-PLM-FAD model, post-training. The ESM-2 embeddings were derived from the pre-trained ESM-2 model, while the learned features were obtained by passing the training data through the model with pre-trained weights. Both techniques diminished the complexity of the datasets to two dimensions, facilitating the presentation of the distinction between FAD and non-FAD groups. The resulting scatter plots illustrate the extent to which the msCNN-PLM-FAD enhances class

Table 12
Comparison of performance and parameter efficiency for msCNN and mCNN.

Models	Sensitivity	Specificity	Accuracy	MCC	AUC	Number of params	Reduced params
***ms*CNN**	**0.8704**	**0.9836**	**0.9795**	**0.7491**	0.9614	**5,955,074**	**79,35%**
*m*CNN	0.8704	0.9801	0.9762	0.7224	**0.9668**	28,841,474	0%

Table 13
Comprehensive protein-level performance across standard and newly curated independent datasets.

Protein	Dataset Group	Sensitivity	Specificity	Accuracy	MCC	AUC	Samples
P53571	Newly Constructed	1.0000	0.9577	0.9595	0.7047	0.9853	321
P56968	Newly Constructed	1.0000	0.9526	0.9542	0.6389	0.9943	262
Q16881	Newly Constructed	0.9500	0.9269	0.9276	0.5048	0.9564	649
C6KT68	Standard	0.8500	0.9915	0.9838	0.8415	0.9457	371
O67931	Standard	0.8571	0.9880	0.9837	0.7697	0.9591	430
Q9AL95	Standard	1.0000	0.9812	0.9815	0.7004	0.9857	379
P0DPQ8	Standard	1.0000	0.8723	0.8772	0.4582	0.9808	334
Total		**0.9381**	**0.9521**	**0.9516**	**0.6079**	**0.9661**	**2746**

Table 14
Detailed residue-level predictive precision across seven independent experimental structures at 5.0 Å contact threshold.

Protein	True Contacts	Top-10 Precision	Top-20 Precision	Enrichment Factor
P53571	29	1.00	0.70	11.9x
Q9AL95	38	1.00	0.90	11.9x
O67931	43	0.80	0.75	9.5x
Q16881	48	0.70	0.55	8.3x
P56968	27	0.60	0.60	7.1x
P0DPQ8	28	0.50	0.40	6.0x
C6KT68	18	0.30	0.35	3.6x
Average	33	0.70	0.61	8.3x

separability compared to the raw ESM-2 embeddings. The t-SNE visualizations highlight local structure preservation, while UMAP provides complementary insights by better capturing global data structure and preserving inter-cluster distances. Highlighting the model's ability to extract discriminative features for accurate classifying of FAD binding sites in protein sequences, the results show that the learnt features from msCNN-PLM-FAD create more distinctive clusters for FAD and non-FAD classes in both t-SNE and UMAP representations.

3.12. SHAP analysis for model interpretability

To elucidate the contribution of our model for FAD binding site classification, we employed SHAP (SHapley Additive exPlanations) [30] to quantify feature importance. Using the GradientExplainer from the SHAP library, we computed SHAP values on a subset of the test dataset. Three visualizations were generated to interpret the model's behavior. Fig. 8 displays the top 20 embedding dimensions with the highest mean absolute SHAP values averaged across residues. This plot highlights the most influential features. As shown in Fig. 9, a bar chart illustrates the mean absolute SHAP values across 17 residue positions. These SHAP values quantify the importance of contextual amino acid positions surrounding the FAD candidate site. The plot clearly reveals that specific neighboring residue positions exert a significant influence on the model's classification of FAD-binding sites. We efficiently obtained this important contextual information of FAD binding locations by using the msCNN-PLM-FAD. Choosing an appropriate sliding window to optimally utilize the number of neighboring amino acids to contribute to the FAD classification decision resulted in excellent classification performance, a major improvement over previous methods. Fig. 10 depicts the mean absolute SHAP values across all residue positions and embedding dimensions. The heatmap reveals patterns of feature importance across the input space. These visualizations collectively demonstrate how specific residue positions and embedding dimensions contribute to the model's ability to distinguish FAD binding sites.

3.13. Spatial corroboration of learned features via 3D visualization

To bridge the gap between our 1D sequence-based analysis and its 3D biological relevance, we performed a structural visualization. Per-residue SHAP contribution scores were computed for the entire P53571 protein (PDB: 1O97) from the independent test set. These scores were then normalized and mapped to the B-factor column of the corresponding 3D structure. The result is presented in Fig. 11. The visualization reveals a striking spatial correlation: (1) High-Importance Regions (Red): Residues yielding the highest positive SHAP values, which represent the features most influential for a FAD prediction, are not randomly distributed. Instead, they are precisely clustered at and around the FAD-binding pocket. (2) Contact Sites: These high-

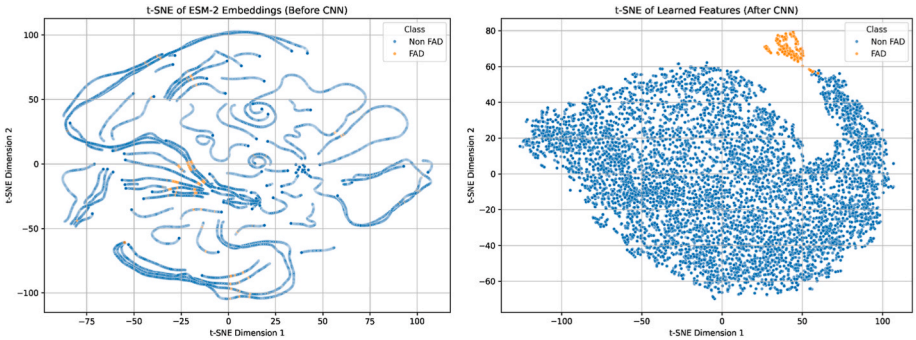


Fig. 6. Visualization of Feature Representations Using t-SNE.

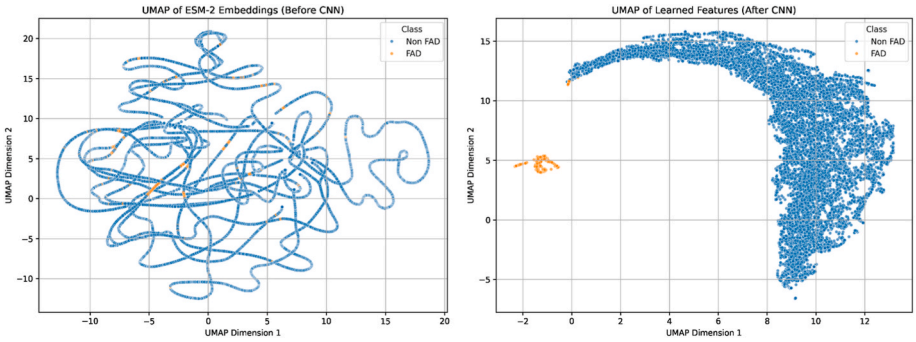


Fig. 7. Visualization of feature representations using UMAP

importance red regions are observed to be in direct spatial contact with the FAD ligand (rendered as green sticks). (3) Background Regions (White): The vast majority of the protein backbone, which constitutes the background with near-zero or slightly negative SHAP scores, is rendered in white.

Using only 1D sequence data, our proposed model has learned sequence patterns that precisely match the 3D spatially-conserved functional site. This study gives significant visual support. Furthermore, this capability transforms the model into a valuable heuristic tool for structural biology. It demonstrates that predictive insights derived purely from 1D sequences can be leveraged to explore the active site in 3D space. This empowers researchers to gain a clearer, spatially-contextualized understanding of the FAD-binding mechanism, allowing for the observation and investigation of potential ligand interactions within the protein's tertiary structure, even in the absence of experimental structural annotations. Finally, this successful mapping highlights a clear pathway for future methodologies to go beyond implicit sequence learning. By directly integrating 3D structural embeddings or employing graph-based architectures to explicitly model the residue-residue spatial proximity observed here, future models could establish a generalized framework for deciphering complex structure-function relationships directly from sequence data, extending the impact well beyond the specific scope of FAD binding.

3.14. *Quantitative Evaluation of Structural Predictive Utility*

To extend the model's interpretability beyond qualitative representations, we conducted a systematic quantitative assessment across a combined dataset of seven proteins from both the standard independent test set (P0DPQ8, C6KT68, O67931, Q9AL95) and the novel non-redundant independent test set (P53571, P56968, Q16881). A primary challenge in validating sequence-based predictions against 3D structures is the frequent indexing mismatch between FASTA sequences and PDB structures, often caused by missing residues in experimental models. To resolve this, we implemented a sequence alignment protocol to establish a rigorous one-to-one residue correspondence between the full-length UniProt sequence (1D) and the experimentally resolved PDB structure (3D). This process explicitly corrects for indexing discrepancies caused by disordered regions or missing residues in the crystallographic models. This ensures that SHAP-based importance scores are accurately projected onto their precise spatial coordinates, enabling a reliable calculation of residue-level precision and recall against experimentally validated contact sites.

The global Precision-Recall analysis (Fig. 12A) demonstrates that residues assigned the highest importance by msCNN-PLM-FAD are significantly concentrated within functional binding pockets. In this context, the Random Baseline represents the prevalence of binding residues within the combined test set, calculated as the ratio of total FAD-contact residues to the total number of residues. For our specific evaluation set, this baseline is approximately 0.084. Despite this low class prevalence, the model achieves a precision of 0.90 in its Top-10 ranked residues (Fig. 12B). This results in an enrichment factor of 10.7x over the random baseline, as shown in the global evaluation (Fig. 12B), providing rigorous evidence of the model's predictive utility in identifying experimental target sites from sequence data alone.

To further detail this performance, Table 14 provides a granular breakdown of predictive precision for each protein in the combined independent set. The stability of these predictions was further validated by analyzing the average precision across the seven individual proteins over a continuous spectrum of distance thresholds from 4.0 to 10.0 Å (Fig. 13). The shaded areas represent the 95% confidence intervals calculated via non-parametric bootstrapping (K = 5000 iterations), consistent with the statistical rigor applied in our primary performance evaluations. While atomic-level precision at the 4.0 Å threshold reflects

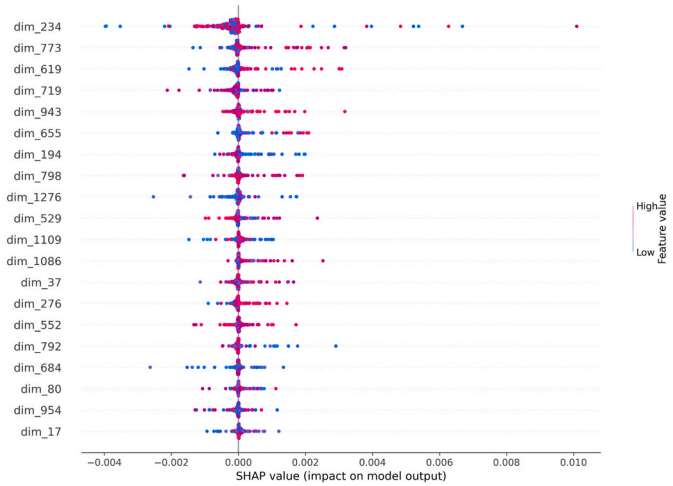


Fig. 8. Summary of SHAP values for top embedding dimensions.

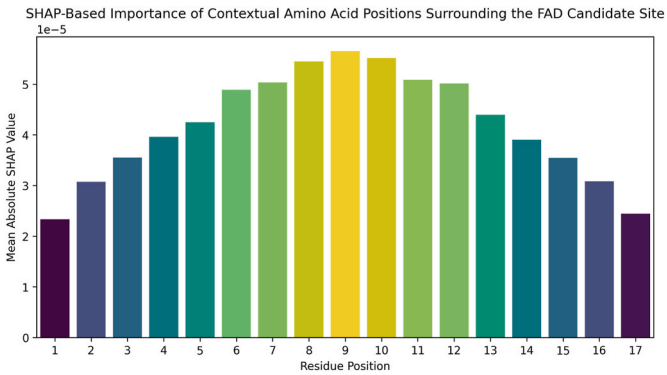


Fig. 9. SHAP values for residue position importance.

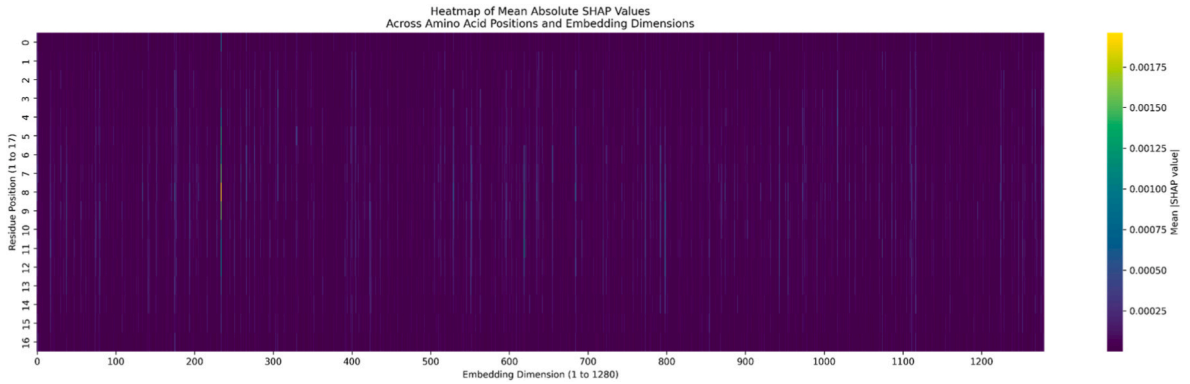


Fig. 10. Heatmap of SHAP values across residues and embeddings.

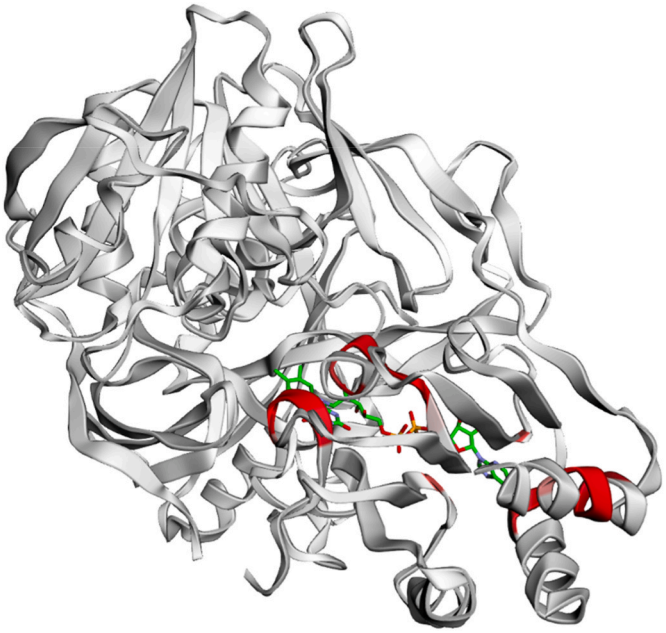


Fig. 11. Spatial mapping of per-residue SHAP importance scores on the P53571 structure (PDB: 1O97).

the inherent structural complexity of diverse protein families, the precision across all proteins converges toward 1.0 as the search radius expands to the pocket neighborhood (6.0–8.0 Å). This trend indicates that high-SHAP residues not in direct contact with the ligand are primarily

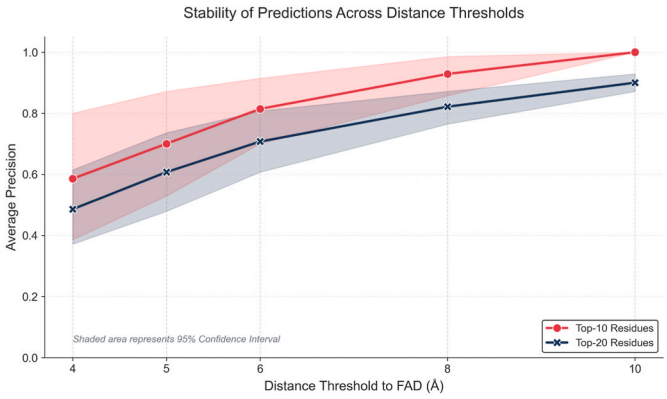


Fig. 13. Stability of predictions across distance thresholds. *The line plot displays the average Precision at Top-10 and Top-20 ranked residues as a function of the distance threshold to the FAD ligand. Shaded areas represent the 95% confidence intervals determined* via *non-parametric bootstrapping (K = 5000). Precision converges to 1.0 as the threshold encompasses the functional neighborhood (>6.0 Å), indicating the model effectively captures the spatial environment of the binding pocket.*

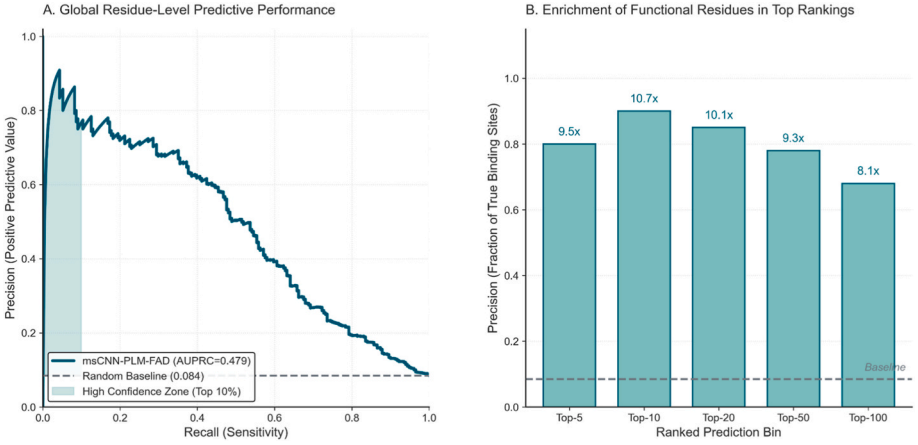


Fig. 12. Systematic evaluation of residue-level predictive utility. *(A) Global Precision-Recall curve aggregated across seven independent test proteins. The shaded region identifies the high-confidence prediction zone where precision remains significantly above the random baseline (0.084), which reflects the actual binding site prevalence in the test set. (B) Enrichment of true FAD-binding residues within ranked prediction bins. The model achieves an enrichment factor of 10.7x for the Top-10 ranked residues, demonstrating its effectiveness in screening candidate sites for structural studies.*

second-shell residues residing within the immediate spatial proximity of the binding pocket. These residues likely contribute to the functional micro-environment or structural scaffolding required for ligand docking. While the enrichment factor reaches a peak of 11.9x for specific proteins like P53571, the consistent average enrichment of 8.3x across all tested structures (Table 14) establishes msCNN-PLM-FAD as a reliable tool for characterizing the functional footprint of FAD binding sites in 3D space.

3.15. *Visualize significant motif inside msCNN-PLM-FAD*

To enhance our understanding and identify the most unique motifs within the survey dataset, we initially deconstructed the well-trained msCNN-PLM-FAD and extracted solely the outputs from the 1-max pooling layers. We employed the Random Forest classifier, an ensemble learning technique primarily grounded in decision trees, to rank all preserved filters from the independent set. We subsequently chose the highest-ranking filters for the display. In the following stage, multiple FAD-binding samples from the testing set were fed into the model for prediction, and the sites where the convolution kernels identified the most promising motifs based on evolutionary information were recorded. Fig. 14 depicts the relationship between evolutionary motifs and their respective filters. We can observe the correlation between the motifs and the filters at the representative position 1074 in the embeddings of FAD binding sites, specifically at positions 105 and 435 in the protein sequence with UniProt ID O67931. This is similar to how the original CNN model detects object features in image data. Rather than depending on one window, the suggested model used several windows to investigate and capture motifs. This method increased the effectiveness of pattern recognition and enabled the identification of motifs of different sizes across the protein sequence. In the examples provided, similarities are consistently observed within specific regions, even among different proteins. Overall, the application of convolutional kernel scans with multiple lengths on the protein evolutionary information led to enhanced protein recognition performance.

4. Conclusion

In this study, we propose a new computational model named *ms*CNN-PLM-FAD to be applied to the FAD binding sites classification problem. By integrating using protein language representation combined with multiple separable windows convolutional neural networks, our model achieves a high AUC result (0.9614), accompanied by a high MCC value (0.7491) showing high discriminative precision and a robust balance between classes, with a sensitivity of 0.8704, specificity of 0.9836, and accuracy of 0.9795. This predictive ability is superior to the method proposed in the previous study, especially for minimizing false positives, highlighting the contribution of our proposed model in both fundamental research and practical drug discovery associated with FAD binding sites.

Simultaneously, our results also demonstrate the effectiveness of pretrained protein language models from the ProtTrans and ESM families in encoding protein sequences into embedding representations especially when combined with multiple separable windows convolutional neural networks. The complex characteristics of the FAD binding sites

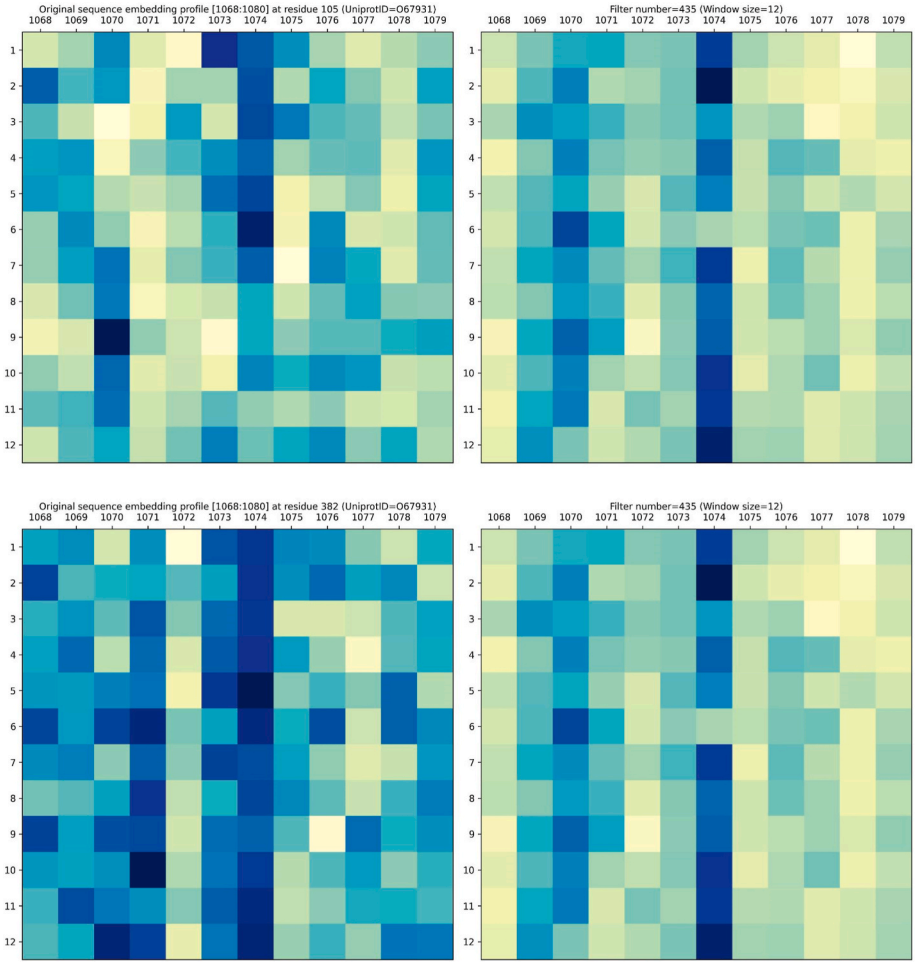


Fig. 14. Visualization of the significant motif and its corresponding position in the chain, with the most characteristic motif identified (right) based on importance ranking using the Forest Ensemble method.

classification problem are captured, overcoming the limitations of previously used conventional techniques such as PSSM or traditional CNN.

The approach suggested in our research not only provides insightful analysis of the capacity to categorize FAD binding sites but also sets the way for use in other protein-related issues, hence laying a basis for numerous future applications in biological knowledge and medication development. The code and data for this study are publicly available at: https://github.com/kioenguyen0207/msCNN-PLM-FAD.

CRediT authorship contribution statement

Dinh-Quy Nguyen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Viet-Thanh Nguyen:** Validation, Software, Resources, Conceptualization. **Cam-Hong Ly:** Data curation, Conceptualization. **Yu-Yen Ou:** Supervision. **Quang-Thai Ho:** Supervision, Software, Resources, Methodology, Investigation.

Additional files

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ab.2026.116092.

Data availability

The code has been shared and referenced in the manuscript.

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